

Credit Constraints and the Missing Recovery

Homebuilders and Housing Supply

Augustus Kmetz

Stanford University

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- 1 Motivation and Introduction
- 2 Builder size affects operations
- 3 Builder size matters for market structure
- 4 Entry has fallen
- 5 Model Ingredients

- Housing affordability is a major concern in the United States, as home prices have generally risen more than household incomes.
- Real prices grew similarly to the 2000's housing boom, but new production was just over half of the earlier boom's.
- There is a bottleneck in housing supply.
- **Research Question:** What is the market structure of new housing and how does it change before and after the bust?

[Buy](#) [Rent](#) [Sell](#) [Get a mortgage](#) [Find an Agent](#)

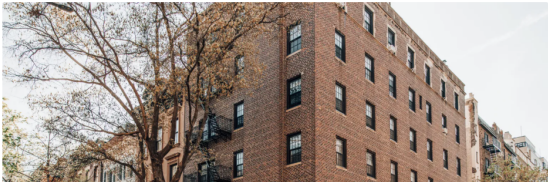


[Zillow Research](#) [Data](#) [Market Trends](#) [Renters](#) [Affordability](#) [Climate Risk](#) [Policy & Industry](#) [Dash](#)

NEWS

In 233 U.S. Cities, Even a Starter Home Costs \$1 Million

Half of all states have at least one city where entry-level homes top \$1 million — a lasting effect of pandemic-era price growth.



- Housing supply is typically studied through the lens of supply elasticities, which relate quantity growth to price changes,
 - These elasticities are estimated using reduced-form and structural methods, and fell after the Great Recession.
 - Many of these elasticities explain little about the past 25 years (Louie et al., 2025)
- Little is known about the mechanics of residential construction in the United States, and understanding the source of disruption to housing markets has large welfare implications.
- **This presentation:** Describe the business of residential construction and present new stylized facts on residential construction in the United States.
 - Large builders build cheaper, more similar homes.
 - Large builders dominate the aggregate production of single-family homes. Small firms contracted during the crash, and broadly did not recover to 2000 levels.
 - Entry has fallen, even with consistent price increases.
 - These patterns are consistent with credit constraints, which bind small firms more.
- **Future:** Develop a model with developer and geographic heterogeneity.

- **Housing Supply:** Housing supply (Dipasquale, 1999), house price changes (Mayer and Somerville, 2000b), land constraints (Saiz, 2010), regulatory barriers (Mayer and Somerville, 2000a) (Gyourko, Saiz, and Summers, 2008) (Gyourko, Hartley, and Krimmel, 2021), elasticities (Aastveit, Albuquerque, and Anundsen, 2023) (Baum-Snow and Han, 2024), the closing of the frontier (Glaeser and Gyourko, 2025), issues with elasticities (Davidoff, 2013) (Louie, Mondragon, and Wieland, 2025)
 - **Contribution:** Describe the market structure of residential construction.
- **The Great Recession and Housing:** Competition (Quintero, 2022), lower elasticities and regulation (Aastveit et al., 2023), labor supply shocks (Le, 2025), large firms (van Straelen, 2023), and credit frictions (Gabrovski and Ortego-Martí, 2025)
 - **Contribution:** Show shifts in the market structure of residential construction.
- **Macro-Housing and Credit:** Housing IS the business cycle (Leamer, 2007), loose credit (Justiniano, Primiceri, and Tambalotti, 2019), credit matters for small firms (Chodorow-Reich, 2013) (Ayes and Raveendranathan, 2023) (Gabrovski and Ortego-Martí, 2025)
 - **Contribution:** Provide evidence consistent with credit constraints mattering for small firms.

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Speculative builders' activities vary by size

- Most new single-family homes in the US are built by “speculative builders.” Speculative builders build homes before a buyer is secured.
- Builders need land, labor, and financing to build homes.
- Builders operations differ by size.

Small firms:

- Build 1-25 homes in a year
- 97% of active firms in 2019
- Build a variety of houses on smaller lots
- Land is bought before the construction loan is acquired
- Finance with short-term construction loans or cash
- Typically do not have a mortgage arm, and prefer cash and conventional loans as payment

Large firms:

- Build more than 100 homes in a year
- <1% of active firms in 2019
- Build homes from standard plans on large tracts
- Land is acquired either through purchases or land options
- Finance with lines of credit, homebuilder notes, or cash
- Typically can issue mortgages through a financing arm, and take mortgages as payment

Large builders build similar, cheaper homes

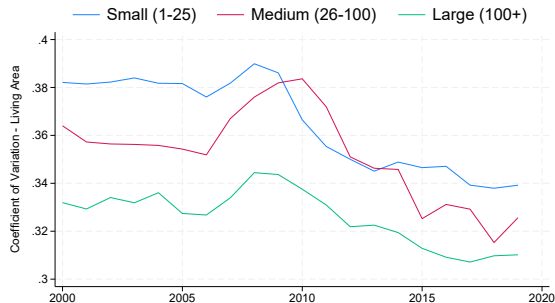


Figure: Coefficient of Variation of Living Area by Firm Size

Notes: This plot shows the coefficient of variation by firm size category over time.

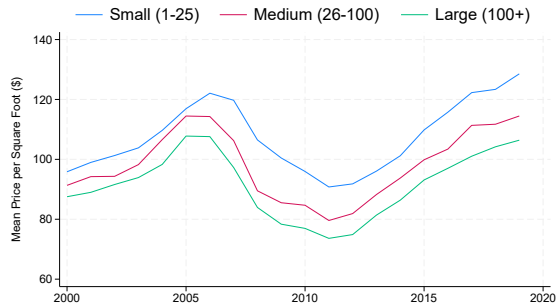


Figure: Mean Price per Square Foot by Firm Size

Notes: This plot shows the mean real price per square foot by firm size category over time. Adjusted to real terms using CPI excluding shelter.

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Large builders dominate the aggregate, but their market share varies in the cross-section

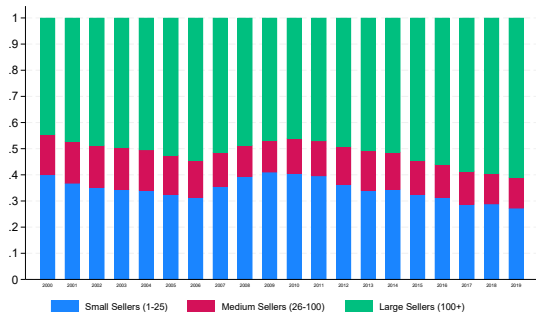


Figure: Aggregate Market Shares by Size

Notes: This plot shows the number of homes produced by each firm-size segment in each year using Cotality data.

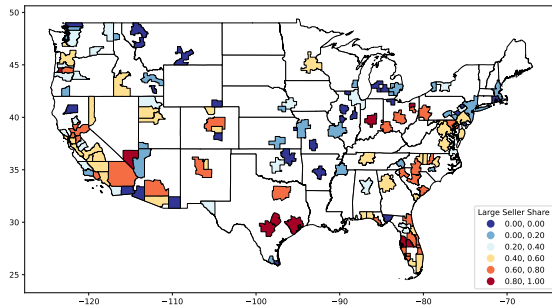


Figure: Large Builder Market Share (2019)

Notes: This map shows the share of homes sold by the large sellers for observed CBSAs in 2019 using Cotality data.

In aggregate, large builders recover and small builders do not



Figure: Relative Production by Firm Size

Notes: This plot shows the relative number of homes (2000=1) produced by each firm-size segment in each year using Cotality data.

This pattern is particularly strong for the largest 10 and 100 firms

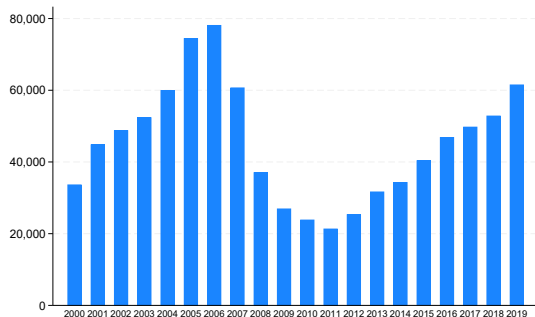


Figure: Top 10 Home Counts

Notes: This plot shows the number of homes sold by the 10 largest builders active within a calendar year.

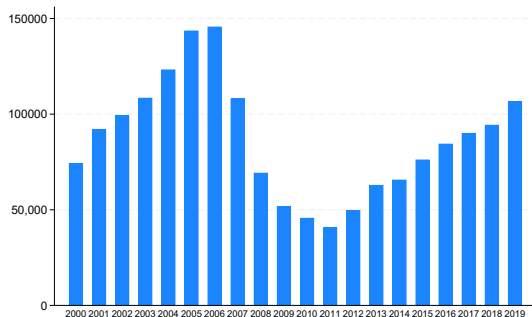


Figure: Top 100 Home Counts

Notes: This plot shows the number of homes sold by the 100 largest builders active within a calendar year.

►► Top 10/100 Share Plots

In the cross-section, large sellers mostly recover, but small sellers do not

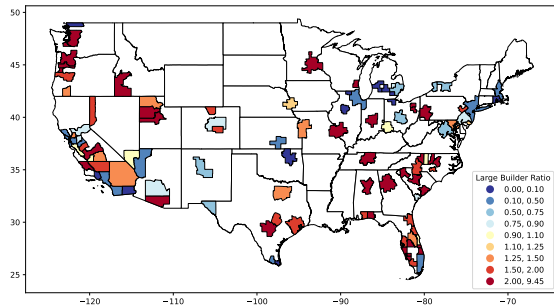


Figure: Relative Production for Large Builders

Notes: This plot shows the production by large firms in 2019 relative to 2000 using Cotality data.

► Relative Production Distribution

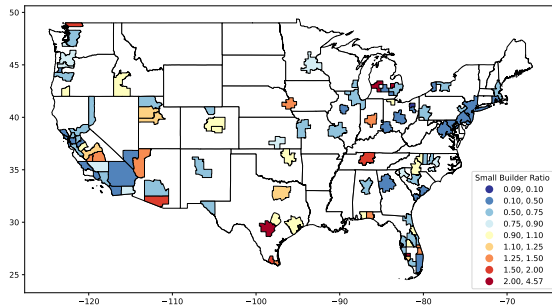


Figure: Relative Production for Small Builders

Notes: This plot shows the production by small firms in 2019 relative to 2000 using Cotality data.

Case Study: Akron vs Providence

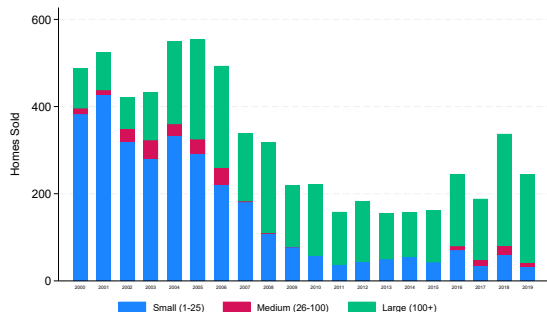


Figure: Akron, OH Production

Notes: This plot shows the number of homes produced by each firm-size segment in Akron, OH by year using Cotality data.

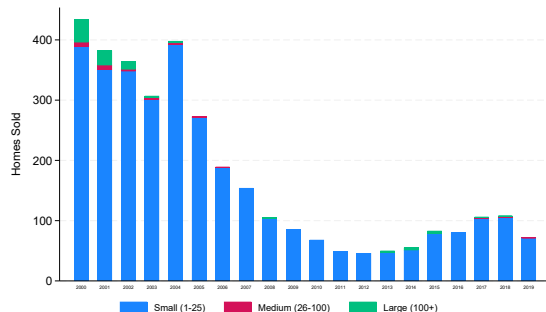


Figure: Providence, RI Production

Notes: This plot shows the number of homes produced by each firm-size segment in Providence, RI by year using Cotality data.

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Entry has fallen at the extensive and intensive margins

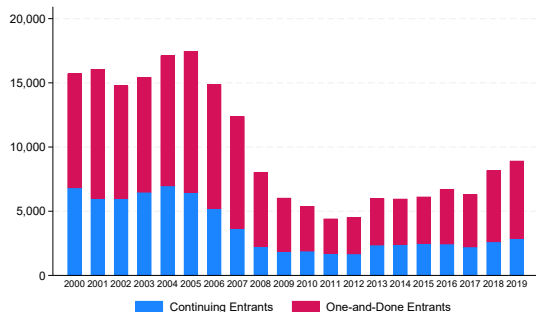


Figure: Number of Entrant Builders Over Time

Notes: This plot shows the number of entrant firms in each year of the Cotality built-for-sale data set. Continuing entrants are entrants who do not exit in their first year of operation. One-and-done entrants exit the same year they enter.

►► Aggregate Continuing Entrants

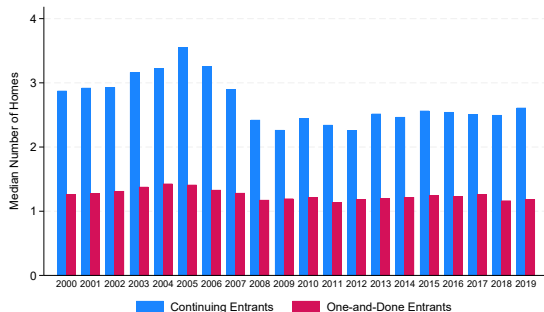


Figure: Average Number of Homes per Entrant Builder Over Time

Notes: This plot shows the average number of homes built by entrants by continuation status in the Cotality built-for-sale data set.

This decline in entry is widespread, and is even worse for continuing entrants

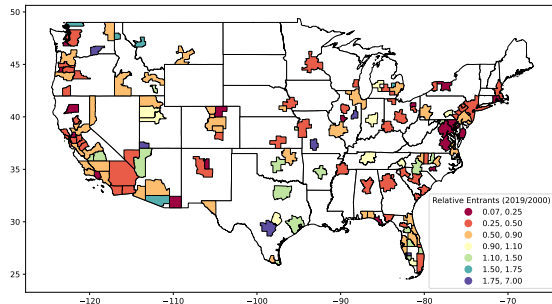


Figure: Relative Entry (2019)

Notes: This map shows the relative number of entrant firms in 2019 relative to 2000 in the Cotality built-for-sale data set.

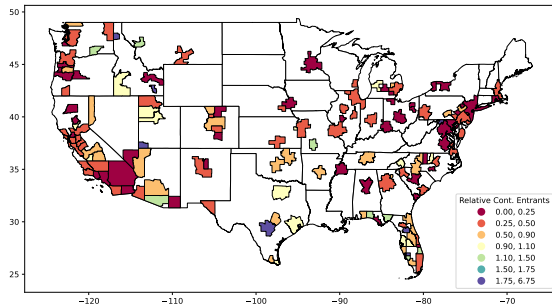


Figure: Relative Continuing Entry (2019)

Notes: This map shows the relative number of continuing entrant firms in 2019 relative to 2000 in the Cotality built-for-sale data set.

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- Two builder types, one with a credit constraint and the other who needs to individually search for funding.
 - Can have a representative large builder who competes with a continuum of small builders
 - Competition is key here, likely imperfect depending on size of small firm market.
- Large builders build few types of home size at a set home quality, and small builders build a range of sizes and qualities.
- High- and low-income consumers purchase homes subject to search frictions
- **Need to match:**
 - Firm response to price changes
 - Relative recovery of large firms versus small firms
 - Depressed entry of small firms
 - Convergence in housing quality $\implies$ could generate spillovers as high income may occupy homes typically used by lower income because they have a strong preference for ownership
 - Overflow goes to rent in apartments, beyond scope of this project.

Thank you! Please let me know if you have any suggestions.

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Summary Statistics

	Mean	50th percentile	Standard deviation	Minimum value	Maximum value
living_area	2408.017	2258	833.1466	992	6195
number_bedrooms	3.578742	4	.8010869	1	6
number_bathrooms	2.911725	3	.8054943	1	6
sale_amount_real	238732.9	206258.2	126852.6	31569.56	1233250
ppsf	100.1006	90.74078	41.50909	7.807833	908.7827
price_per_bedroom	67904.46	58649.27	36072.09	6848.796	1152293

Table: Building Facts Summary Statistics

Notes: This table shows descriptive statistics for the variables examined in this section. These data use the Cotality new construction data as the base, but keep observations that are not missing living area, the number of bedrooms, the number of bathrooms, and the price per square foot.

Large firms are cheaper even controlling for CBSA and year

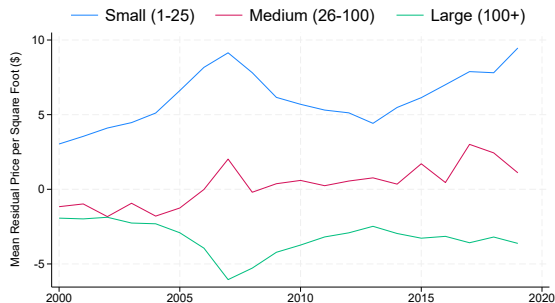


Figure: Mean Residual Price per Square Foot by Firm Size

Notes: This plot shows the mean residual price per square foot (location-adjusted) by firm size category over time.

[▶▶ Back to Builder Facts](#)

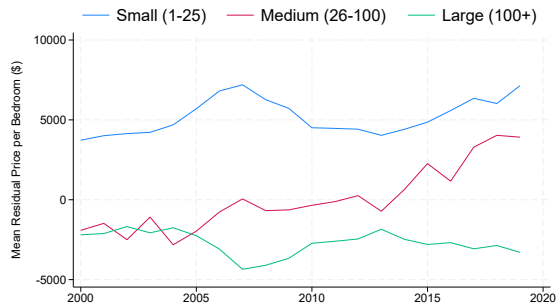


Figure: Mean Residual Price per Bedroom by Firm Size

Notes: This plot shows the mean residual price per bedroom (location-adjusted) by firm size category over time.

Large and small builders customer bases are different

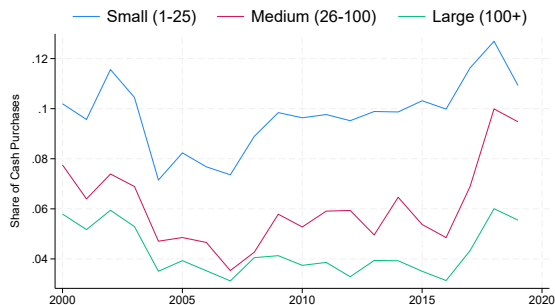


Figure: Share of Cash Purchases by Firm Size

Notes: This plot shows the share of cash purchases by firm size category over time.

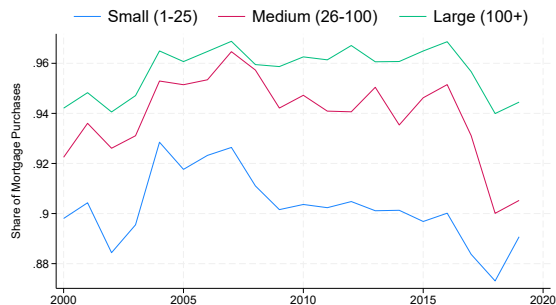


Figure: Share of Mortgage Purchases by Firm Size

Notes: This plot shows the share of mortgage purchases by firm size category over time.

► Back to Builder Facts

Large and small builders take different mortgage products

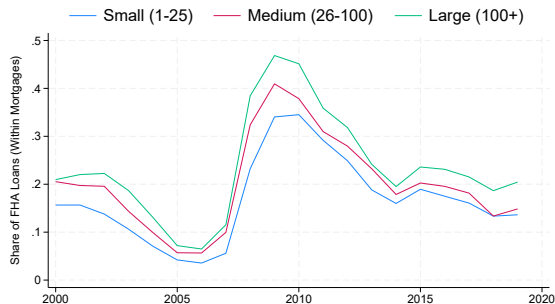


Figure: Share of FHA Loans by Firm Size Within Mortgages

Notes: This plot shows the share of FHA loans by firm size category over time, conditional on the purchase being financed by a mortgage.

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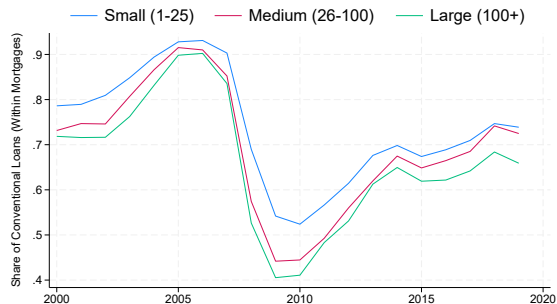


Figure: Share of Conventional Loans by Firm Size Within Mortgages

Notes: This plot shows the share of conventional loans by firm size category over time, conditional on the purchase being financed by a mortgage.

Market Shares by Size

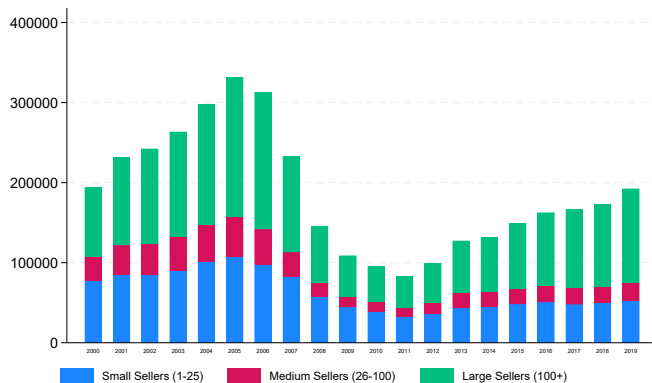


Figure: Home Production by Firm Size

Notes: This plot shows the number of homes produced by each firm-size segment in each year using Cotality data.

Market Shares by Size

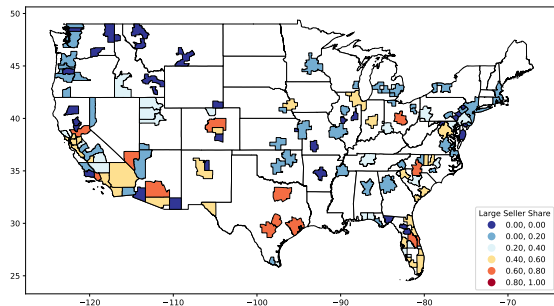


Figure: Large Builder Market Share (2000)

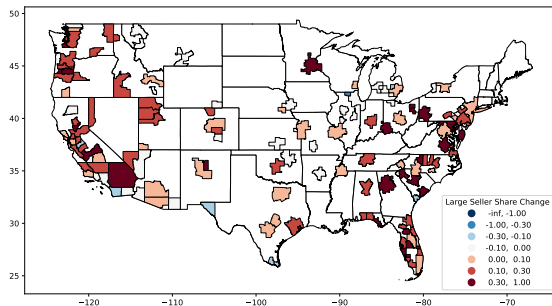
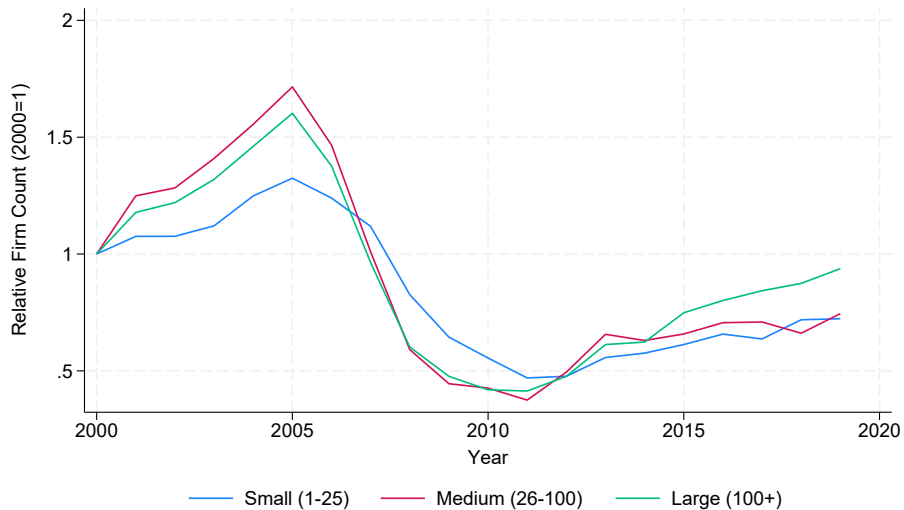


Figure: Large Builder Market Share Change (2000-2019)

► Back to Large Sellers

There was aggregate firm destruction



This pattern is particularly strong for the largest 10 and 100 firms

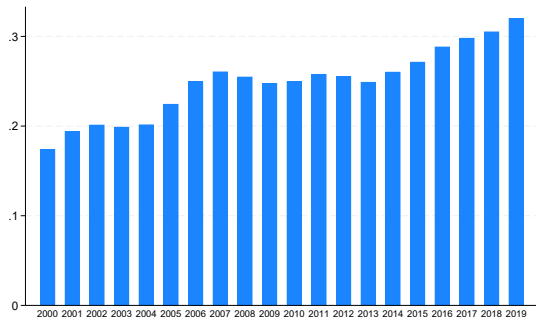


Figure: Top 10 Home Share

Notes: This plot shows the share of homes sold by the 10 largest builders active within a calendar year.

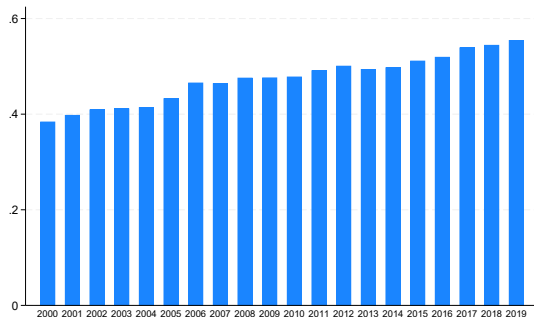


Figure: Top 100 Home Share

Notes: This plot shows the share of homes sold by the 100 largest builders active within a calendar year.

[▶▶ Back to Top 10/100 Share Plots](#)

In the cross-section, large builders mostly recover but small builders do not



Figure: Relative Production for Large Builders

Notes: This plot shows the production by large firms in 2019 relative to 2000 using Cotality data.



Figure: Relative Production for Small Builders

Notes: This plot shows the production by small firms in 2019 relative to 2000 using Cotality data.

[▶▶ Back to 2019 Maps](#)

Continuing entry has remained depressed

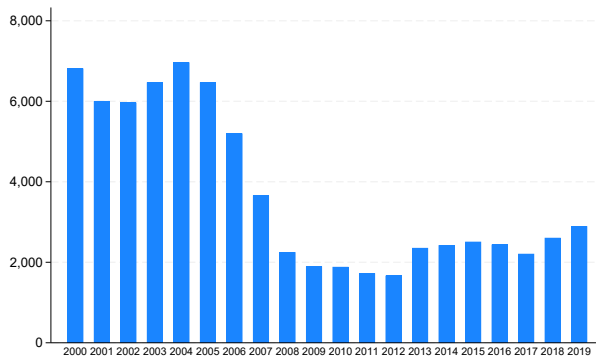


Figure: Number of Continuing Entrant Firms Over Time

Notes: This plot shows the number of homes built by entrants in the Cotality built-for-sale data set. This excludes firms who enter and exit in the same year.

The distribution of relative entry is depressed

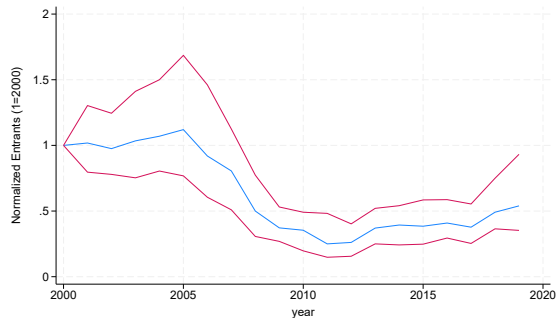


Figure: Relative Entry

Notes: This plot shows the CBSA median and interquartile range of entrants relative 2000 in the Cotality built-for-sale data set.



Figure: Relative Continuing Entry

Notes: This plot shows the CBSA median and interquartile range of continuing entrants relative 2000 in the Cotality built-for-sale data set.